User Story 1

Learning SQL Fundamentals

User Story

As an aspiring SQL Server DBA, I want to learn SQL fundamentals so that I can query and manage data effectively within a database environment.

Tasks

- Research the role and responsibilities of a SQL Server DBA.
- Install SQL Server Developer Edition.
- Install SQL Server Management Studio (SSMS).
- Connect to a local SQL Server instance.
- Learn database concepts (tables, rows, columns, schemas).
- Practice SELECT statements.
- Practice filtering data using WHERE clauses.
- Learn sorting using ORDER BY.
- Practice aggregate functions (COUNT, SUM, AVG, MAX, MIN).
- Complete SQL exercises and sample queries.
- Document lessons learned.

User Story 2

Learning T-SQL and Database Administration Basics

User Story

As an aspiring SQL Server DBA, I want to learn Transact-SQL (T-SQL) and SQL Server administration basics so that I can perform common administrative and troubleshooting tasks.

Tasks

- Learn T-SQL syntax and scripting conventions.
- Create databases using T-SQL.
- Create and modify tables.
- Insert, update, and delete records.
- Learn stored procedures.
- Learn views and functions.
- Practice joins and subqueries.

- Study SQL Server security concepts.
- Learn login and user management.
- Document common DBA scripts.

User Story 3

Monitoring SQL Server Health

User Story

As a SQL Server DBA trainee, I want to monitor SQL Server health metrics so that I can identify performance issues proactively.

Tasks

- Research SQL Server performance monitoring.
- Learn Dynamic Management Views (DMVs).
- Study CPU utilization metrics.
- Study memory utilization metrics.
- Research wait statistics.
- Learn blocking and deadlock concepts.
- Implement WhoIsActive monitoring script.
- Create scripts to detect long-running queries.
- Test monitoring queries in SSMS.
- Save monitoring scripts for dashboard integration.

User Story 4

Designing the SQL Server Dashboard

User Story

As a SQL Server DBA, I want to design a SQL Server Health Dashboard so that key performance indicators can be viewed in a single location.

Tasks

- Define dashboard objectives.
- Identify required KPIs.
- Design dashboard layout.
- List database metrics to display.
- Define API requirements.

- Create solution architecture diagram.
- Plan backend structure.
- Plan frontend structure.
- Define reporting requirements.
- Create dashboard wireframes.

User Story 5

Building the Backend Services

User Story

As a SQL Server DBA, I want to build backend services that collect and expose SQL Server metrics so that the dashboard can display real-time information.

Tasks

- Create Spring Boot project.
- Configure SQL Server database connection.
- Configure application properties.
- Create project package structure.
- Create entity classes.
- Create repository classes.
- Create service layer.
- Create REST API endpoints.
- Test endpoints using Swagger.
- Implement health check endpoints.
- Expose blocking session metrics.
- Expose failed SQL Agent jobs data.
- Expose backup status data.
- Expose wait statistics data.
- Perform backend testing.

User Story 6

Building the Dashboard Frontend

User Story

As a SQL Server DBA, I want a visual dashboard that displays SQL Server health metrics so that performance issues can be identified quickly.

Tasks

- Create frontend application.
- Design dashboard navigation.
- Create health summary cards.
- Create Query Execution Time widget.
- Create Throughput widget.
- Create Latency widget.
- Create Resource Utilization widget.
- Create Failed Jobs widget.
- Create Backup Status widget.
- Create Blocking Sessions widget.
- Create Wait Statistics widget.
- Connect frontend to backend APIs.
- Test dashboard responsiveness.
- Perform UI refinements.

User Story 7

Testing, Troubleshooting, and Project Completion

User Story

As a SQL Server DBA, I want to test and optimize the dashboard so that it provides accurate and reliable database monitoring information.

Tasks

- Perform end-to-end testing.
- Verify all API integrations.
- Validate SQL monitoring scripts.
- Troubleshoot backend errors.
- Troubleshoot frontend issues.
- Optimize SQL queries.
- Improve dashboard performance.
- Test dashboard with real SQL Server data.
- Prepare project documentation.

- Create deployment guide.
- Document lessons learned.
- Prepare portfolio screenshots.
- Create final project presentation.
- Conduct project review and sign-off.